Radiation stress under finite observer boosts and response-preserving energy moments

Anonymous research draft
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A low temperature quadrupole approximates radiation stress only within its amplitude and spectral assumptions. We compare exact bolometric fourth-power conversion with a linear temperature-to-moment adapter on the same six positive synthetic skies and the same correlated errors. Both adapters use the same finite-boost stress residual and invariant rest-pressure anisotropy. Their physical-target coverage and interval widths differ even when their method-target calibration is similar. Separately, energy moments of angular ranks zero, two and four preserve the complete five-dimensional radiative shear-response operator. In a retained compression example, rank-four response agreement reaches numerical precision while a positive angular entropy residual remains. These are conditional full-sky radiation methods; applying them to measured channels requires calibrated intensity, spectral completion and a measured joint error law. The accompanying finite-bandpass foreground analysis addresses a different inverse problem and does not supply those inputs.

I. PHYSICAL MOMENTS AND TWO COMMON-INPUT ADAPTERS

Use signature $(-+++)$, outward sky direction n, photon direction $e=-n$, and bolometric energy density per solid angle I in J m^{-3} sr^{-1}. With a calibrated energy temperature,

$$I = C_\gamma T_E^4, \quad C_\gamma = \frac{\pi k_B^4}{60\hbar^3 c^3}, \quad \rho = \int I\, d\Omega, \quad q = -c\int In\, d\Omega, \quad \pi = \int I\,\mathrm{STF}(nn)\, d\Omega.$$

The reference temperature is 2.7255 K, giving $\rho_{\rm ref} = 4.17480092\times10^{-14}$ J/m^3. Set $f = q/(c\rho)$, $P=\pi/\rho$. On the positive moment domain with a unique timelike Landau eigenbranch, define

$$\mathscr{D} = P - \frac{3\,\mathrm{STF}(ff^T)}{2+\sqrt{4-3|f|^2}}, \qquad \mathcal{A}_\gamma = \frac{[\sum_{i=1}^3 (p_{L,i}-\rho_L/3)^2]^{1/2}}{\rho_L}.$$

The residual $\mathscr{D}$ is an observer-frame STF tensor; $\mathcal{A}_\gamma$ is invariant under observer tetrad changes on this branch. A zero residual tests a timelike isotropic-pressure rest stress. It does not require every higher angular multipole to vanish.

The source brightness is $I_{\rm rest}/I_{\rm ref} = 1 + aP_2(n_{{\rm rest},z})$, with $a\in\{0, 2\times10^{-5}, 0.02\}$, boosted by $\boldsymbol{\beta} = b(0.6,0,0.8)$, $b\in\{0.00123, 0.3\}$. The complete positive temperature sky is obtained from this specified boost, retaining its higher modes. The analytic target is the Lorentz transform of $\mathrm{diag}(1, 1/3-a/15, 1/3-a/15, 1/3+2a/15)\rho_{\rm ref}$, so $\mathcal{A}_{\gamma,\rm true} = \sqrt{6}|a|/15$.

Only the three temperature dipole and five temperature STF coordinates are perturbed. Their Gaussian standard deviations are 7.5×10^{-6} and 9.375×10^{-7}, and their correlation is $0.8I_8 + 0.2ss^T$, $s=(1,-1,1,1,-1,1,-1,1)$. Monopole and higher temperature modes are noiseless known inputs. Both methods receive identical errors. The exact adapter integrates the fourth power of that sky. The linear adapter uses $f_{\rm lin} = -4d_T/3$ and $P_{\rm lin} = 8Q_T/15$, with the same monopole normalization, then applies the same exact $\mathscr{D}$ and Landau readout. This isolates the temperature-to-moment approximation. The joint flux/stress covariance, including its three-by-five cross block, is retained; it is not replaced by independent dipole and quadrupole errors.

II. CONDITIONAL CALIBRATION, COVERAGE AND WIDTHS

Each case uses 8,191 calibration skies and 4,096 disjoint test skies. Calibration is pointwise around the known noiseless response of each method. For $\mathscr{D}$, the confidence-region radius is the 7,783rd ordered calibration norm error, producing a five-dimensional ball. For $\mathcal{A}_\gamma$, the interval subtracts the 7,988th and 204th ordered calibration errors and intersects the result with $[0,\infty)$. An upper endpoint below zero produces an empty interval and is counted as noncoverage. The exact physical target and the noiseless method target are assessed separately; no bias correction is fitted to force them to coincide.

For the pure-boost case at $b = 0.00123$, the exact and linear physical-target $\mathscr{D}$ counts are 3,872/4,096 and 3,035/4,096, while the invariant-anisotropy counts are 3,914/4,096 and 1,545/4,096. Their $\mathscr{D}$ ball diameters are nearly equal, 3.390×10^{-6} and 3.381×10^{-6}. Thus similar widths can conceal different target bias. The linear method-target coverage remains near its nominal level across the cases even where its physical-target coverage fails.

TABLE I. Physical-target hit counts and finite-test 95-percent Clopper–Pearson intervals. $N = 4096$ for every row. Calibration uses 8191 separate draws; calibration-content uncertainty is distinct from these test intervals.

Case	Method	$\mathscr{D}$ hits/N [interval]	$\mathcal{A}_\gamma$ hits/N [interval]
a0_b0.00123	exact	3872/4096 [0.9379,0.9521]	3914/4096 [0.9488,0.9617]
a0_b0.00123	linear	3035/4096 [0.7273,0.7543]	1545/4096 [0.3623,0.3922]
a0_b0.3	exact	3934/4096 [0.9540,0.9662]	3907/4096 [0.9470,0.9601]
a0_b0.3	linear	0/4096 [0.0000,0.0009]	0/4096 [0.0000,0.0009]
a2e-05_b0.00123	exact	3890/4096 [0.9426,0.9562]	3862/4096 [0.9353,0.9498]
a2e-05_b0.00123	linear	3055/4096 [0.7322,0.7591]	3654/4096 [0.8822,0.9014]
a2e-05_b0.3	exact	3919/4096 [0.9501,0.9628]	3892/4096 [0.9431,0.9567]
a2e-05_b0.3	linear	0/4096 [0.0000,0.0009]	0/4096 [0.0000,0.0009]
a0.02_b0.00123	exact	3888/4096 [0.9420,0.9557]	3905/4096 [0.9465,0.9596]
a0.02_b0.00123	linear	0/4096 [0.0000,0.0009]	0/4096 [0.0000,0.0009]
a0.02_b0.3	exact	3873/4096 [0.9382,0.9523]	3901/4096 [0.9454,0.9587]
a0.02_b0.3	linear	0/4096 [0.0000,0.0009]	0/4096 [0.0000,0.0009]

TABLE II. Dimensionless widths displayed in units of 10^{-6}. Empty scalar intervals count as noncoverage; only the explicitly counted nonempty intervals enter their mean.

Case	Method	Ball diameter $/10^{-6}$	Empty fraction	N_{nonempty}	Mean width $/10^{-6}$
a0_b0.00123	exact	3.390224	0.021729	4007	0.638565
a0_b0.00123	linear	3.380536	0.000000	4096	1.490733
a0_b0.3	exact	8.369104	0.029053	3977	1.884719
a0_b0.3	linear	27.694794	0.000000	4096	33.419948
a2e-05_b0.00123	exact	3.396305	0.000000	4096	1.887453
a2e-05_b0.00123	linear	3.388694	0.000000	4096	1.932102
a2e-05_b0.3	exact	8.454129	0.000000	4096	5.066372
a2e-05_b0.3	linear	27.748430	0.000000	4096	33.723377
a0.02_b0.00123	exact	3.415737	0.000000	4096	1.990204
a0.02_b0.00123	linear	3.420747	0.000000	4096	1.995023
a0.02_b0.3	exact	8.441110	0.000000	4096	6.450267
a0.02_b0.3	linear	27.647888	0.000000	4096	33.857972

At $a = 0, b = 0.3$, the exact $\mathscr{D}$ count is 3,934/4,096, or 0.960449, with a retained 95-percent interval [0.954021, 0.966209]. This single high estimate is an unadjusted finite-simulation diagnostic among the reported comparisons, not a claim of exact 95-percent coverage. Calibration-content uncertainty is separate: under continuous errors its retained central intervals are [0.945254,0.954684] for the ball and [0.945382,0.954801] for the scalar interval. No seeds, quantiles or sample counts were adjusted. Empty exact-method scalar intervals occur in 89 and 119 test skies in the two pure-boost cases; corresponding nonempty means use 4,007 and 3,977 skies. Empirical measured-sky error-law misspecification is outside these intervals.

III. THE MOMENTS NEEDED FOR THE TARGET RESPONSE

Let $H_4 = \int I(e) e_{\langle i} e_j e_k e_{l\rangle}\, d\Omega$. On the five-dimensional STF space, the instantaneous radiative response is

$$\mathcal{L}_I S = \frac{8\rho}{15} S + \frac{10}{7}\,\mathrm{STF}[\tfrac{1}{2}(S\pi + \pi S)] - H_4 : S,$$

with quadratic form

$$S : \mathcal{L}_I S = \int I(e)[2|Se|^2 - (e^T S e)^2]\, d\Omega.$$

Only energy moments of ranks zero, two and four enter. Matching these moments therefore preserves the entire operator, independently of unresolved higher angular structure. In the retained compression experiment, the operator error normalized by ρ_0 is 0.00654243536 at rank two and 3.3393×10^{-16} at rank four on the primary quadrature, while the positive normalized angular entropy residual is $9.37160903 \times 10^{-5}$. The higher quadrature gives rank-four error 4.3301×10^{-15}; the conclusion is numerical agreement at the checked precision, not exact binary equality.

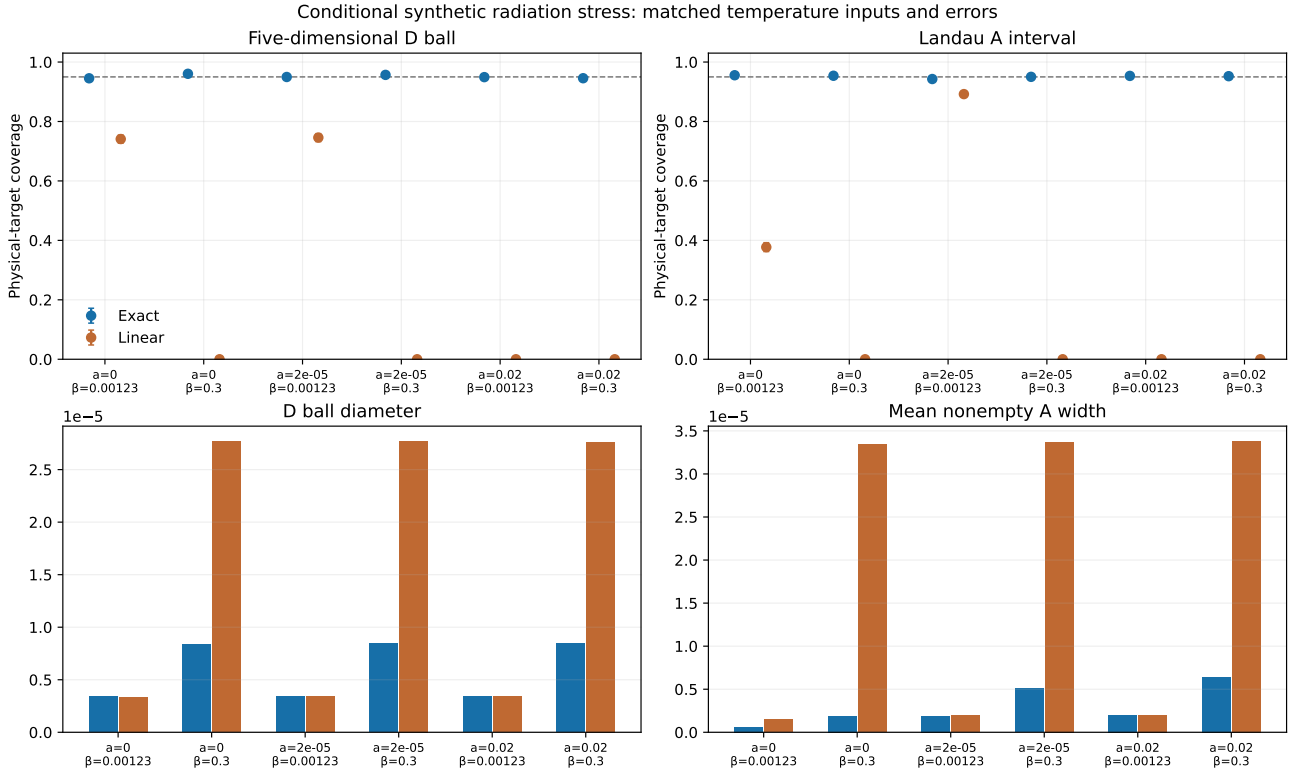


FIG. 1. Common-input synthetic comparison of exact and linear temperature adapters. Upper panels show physical-target coverage and finite-test Clopper–Pearson intervals; the dashed line is the nominal calibration level. Lower panels show the five-dimensional residual-ball diameter and the mean width of nonempty rest-anisotropy intervals. Empty intervals count as noncoverage and are excluded only from the explicitly labeled nonempty width means. Both methods use the same correlated temperature errors, with noiseless known monopole and higher modes. These are pointwise oracle benchmarks, not calibrated intervals for an unknown observed CMB sky.

TABLE III. Both retained angular quadratures and response errors normalized by ρ_0; entropy residuals use the isotropic entropy normalization.

Quadrature	Rank	Operator error	Entropy residual
256	2	0.0065424354	0.00037370961
256	4	3.3393191e-16	9.371609e-05
320	2	0.0065424354	0.00037370961
320	4	4.3301414e-15	9.371609e-05

For the positive $e^{8\mu^2}$ brightness family, a sufficient entropy lower bound is -0.352935ρ, while the actual minimum response eigenvalue is $0.12823282\rho > 0$. A failed sufficient certificate does not establish loss of coercivity; the finite operator spectrum provides the direct check. If a specified transport branch supplies $\mathcal{L}_I\sigma = -R$, a shear estimate also needs the derivative residual R, its frame and its error. That residual is absent here, so no matter shear is inferred from instantaneous brightness alone.

The finite-bandpass foreground problem and this bolometric calculation constrain different objects. A calibrated intensity/spectral completion is needed to carry measured channel amplitudes into the positive energy moments used above. Polarization E is not such an intensity, and evaluating number and energy from one imposed blackbody

TABLE IV. All five rank-four response eigenvalues divided by ρ_0. Full matrices accompany the numerical supplement.

Quadrature	λ_1	λ_2	λ_3	λ_4	λ_5
256	0.5034920635	0.5034920635	0.5526984127	0.5526984127	0.5542857143
320	0.5034920635	0.5034920635	0.5526984127	0.5526984127	0.5542857143

temperature does not create a second independent observed channel. The present contribution is a response-specific stopping criterion and a matched-input stress comparison under a stated synthetic law.

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[1] The source-bound numerical supplement distributed with this manuscript retains the complete moment matrices, common-input comparison statistics and qualification limitations.